

Muskan Dhamuria

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EDUCATION

Singapore Management University

Aug 2024 – May 2028

Bachelor of Science Information Systems – Double Major in Product Development and Cybersecurity

- Activities : SMU Women in Tech, SMU .Hack , UWS STEMentorship, SMU AI Club
- Most Feasible Solution in Hack For Cities Hackathon 2026
- Honourable Mention Fintech Innovator's Hackathon 2026
- Teaching Assistant: IS112 Data Management

WORK EXPERIENCE

SP Group

Jul 2026 – Dec 2026

AI Automation Engineer

- Migrated legacy IQ Bot document-processing workflows to modern AI-powered Document Automation solutions, improving maintainability and scalability.
- Designed and optimized intelligent document-processing pipelines to enhance data extraction accuracy and reduce manual processing effort.
- Developed Agentic AI workflows for intelligent decision-making, task orchestration, and end-to-end business process automation.
- Collaborated with business stakeholders and cross-functional teams to gather requirements, design AI-driven automation solutions, and support deployment.
- Leveraged technologies including Automation Anywhere, Document Automation, OCR, Generative AI, and Agentic AI to streamline enterprise workflows.

Science Centre Singapore

Jan 2022 – Feb 2022

Student Intern

- Developed an interactive 'Young Engineer Trail' and online modules for the 'Young Digital Fabricator' badge program.
- Took full ownership of project delivery following teammate withdrawal, ensuring 100% on-time completion of deliverables.

PORTFOLIO

ESMOS Cloud-Native Healthcare Platform

- Designed and deployed cloud-native enterprise applications on Azure Kubernetes Service (AKS) using Docker, ArgoCD GitOps, and CI/CD pipelines, achieving 99.9% uptime. Implemented TLS-secured ingress routing with NGINX Ingress and cert-manager, container orchestration, persistent storage provisioning, monitoring/log analytics, and automated deployment workflows for scalable microservice-based healthcare systems.
- Technologies: Kubernetes, Docker, Azure AKS, Azure Container Registry, ArgoCD, GitHub Actions, NGINX Ingress, cert-manager, PostgreSQL, Azure Files, Azure Managed Disks

Web Vulnerability Scanner

- Built an automated web vulnerability scanner capable of detecting SQL Injection, reflected XSS, and missing security headers.
- Implemented a BFS-based crawler, risk-based vulnerability scoring, and multi-format reporting (HTML, JSON, CSV) to support security assessments.
- Technologies: Python, Requests, BeautifulSoup, Web Crawling (BFS), Application Security Testing (SQLi, XSS), Security Headers, Data Reporting (JSON, HTML, CSV)

Urbanix - AI-powered urban policy optimization platform

- Implemented multi-objective optimization logic and Pareto frontier analysis to simulate and evaluate urban policy bundles. Developed data ingestion and transformation pipelines for Excel-based urban datasets and real-time visualization dashboards.
- Awarded Most Feasible Solution - Hack For Cities Hackathon 2026
- Technologies: React.js, TypeScript, Tailwind CSS, Recharts, SheetJS, Node.js, Ollama, JavaScript

Koryori Hayashi - Smart Restaurant Intelligence System for local Japanese Restaurant

- Designed and developed a full-stack digital transformation ecosystem integrating online ordering, loyalty management, operational analytics, and queue management systems.

- Implemented a machine learning-powered digital twin to simulate customer demand, inventory usage, and business performance under different operational scenarios, supporting data-driven decision-making.
- Achieved Higher Distinction for the project during coursework.
- Technologies: React.js, TypeScript, Tailwind CSS, Node.js, Supabase, Recharts, Gemini API, Machine Learning, Thompson Sampling, Digital Twin Simulation

SKILLS

- **Programming Languages:** Python, Java, JavaScript, TypeScript, PHP, Bash, SQL
- **Web & Software Development:** React.js, Vue.js, Flask, Node.js, Firebase, HTML, CSS, Bootstrap, Tailwind CSS
- **Cloud & DevOps:** Azure, Docker, Kubernetes, ArgoCD, GitHub Actions
- **Cybersecurity:** Kali Linux, Metasploitable, Web Application Security Testing, Vulnerability Assessment, Linux
- **Databases & Analytics:** MySQL, Pandas, Tableau, Power Query, Matplotlib, Folium, Microsoft Excel
- **Tools & Design:** Figma, Signavio